

Rishabh Baral

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U.S. Citizen | Seeking AI/ML Research & Data Science Internships (Summer 2027)

EDUCATION

Arizona State University

Doctor of Philosophy in Computer Science

Tempe, AZ

Expected May 2030

Arizona State University

Master of Science in Computer Science

Tempe, AZ

May 2026

University of Maryland

Bachelor of Science in Computer Science

College Park, MD

May 2024

RESEARCH PUBLICATIONS

Moneyball with LLMs: Analyzing Tabular Summarization in Sports Narratives

Jan 2026

R. Upadhyay, R. Baral, N. Ahuja*, V. Gupta | Accepted as Findings Paper to 64th ACL, San Diego, CA; Jul 2-7, 2026*

PHANTOM RECALL: When Familiar Puzzles Fool Smart Models

Jan 2026

S. Mukhopadhyay, R. Baral*, N. Mahajan, et al. | ACL Rolling Review (ARR)*

- Benchmarked LLM reasoning via puzzle variations to expose memorization vs. true reasoning.
- Developed automated error classification achieving 96% human agreement.

"The Moon that Looks Like a Boat": Evaluating Visual Similes in T2I Models

May 2026

Y. Luo, R. Baral*, S. Mukhopadhyay, Y. Yang, C. Baral | ACL Rolling Review (ARR)*

- Evaluated state-of-the-art Text-to-Image (T2I) models on their ability to comprehend and generate visual similes, highlighting current limitations in figurative language alignment.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C#, Rust, Kotlin, SQL, OCaml, R, MATLAB

ML Libraries: PyTorch, torchvision, Scikit-learn, NumPy, SciPy, Pandas, OpenCV

ML Methods: Computer Vision, Deep Learning, Semi-Supervised Learning, Multi-Task Learning, Transfer Learning

Architectures: ResNet-18, ResNet-50, U-Net, Vision Transformers (ViTs), RAD-DINO

Medical & GIS: DICOM, pydicom, MIMIC-CXR, MedMNIST, TBX11K, QGIS, ArcGIS, GDAL

Tools: CUDA, Git, MongoDB, ImageMagick, VS Code, Android Studio

EXPERIENCE

Arizona State University

Tempe, AZ

Graduate Project / Teaching Assistant

Aug 2024 – Present

- Sports Narrative Tabular Summarization:** Contributed to the development and evaluation of LLM-based systems for transforming long-form sports commentary into structured statistical tables. Evaluated GPT-4o and Gemini 2.0 Flash on multi-entity reasoning, numerical consistency, and table generation robustness, leading to a Findings of ACL 2026 publication.
- Information Assurance:** Performed a comprehensive security analysis for the healthcare industry by applying AI/ML techniques to identify systemic vulnerabilities. Evaluated information assurance protocols and predictive modeling to ensure data integrity and patient privacy within clinical and multimedia database environments.
- Systems:** Architected and implemented a fully functional Distributed Hash Table (DHT) in Python to manage and retrieve NOAA weather records efficiently. Optimized data distribution, lookup latency, and fault tolerance to handle high-concurrency access to large-scale environmental datasets.
- Instruction:** Serve as a Computer Science Lab Teaching Assistant, leading instructional sessions on JavaScript for undergraduate students. Responsible for grading technical assignments, troubleshooting full-stack code, and delivering comprehensive review guides to reinforce object-oriented design and programming principles.

SELECTED PROJECTS

- Unified 2D/3D Multi-Task Framework for Clinical Radiographs:** Engineered a dimension-agnostic deep learning framework using PyTorch and Axial-Coronal-Sagittal (ACS) convolutions to unify 2D and 3D medical image analysis across 18 MedMNIST v2 datasets. Developed a semi-supervised Student-Teacher pipeline integrating RAD-DINO and Exponential Moving Averages (EMA) to leverage 184,556 unlabeled MIMIC-CXR images, achieving a state-of-the-art Test AUC of 0.878. Implemented an Ark Cyclic Training paradigm within a ResNet-50 backbone to eliminate gradient interference across concurrent classification, localization, and segmentation tasks, while optimizing deployment on Apple M3 silicon to achieve a 2.4× throughput increase. Conducted 20 independent experiments on TBX11K, demonstrating that training from scratch significantly outperformed COCO pre-training in clinical imaging domains.

- **Clinical Data Pipeline & Diagnostic Architectures:** Built a medical imaging pipeline using `pydicom` to parse clinical metadata and normalize high-bit-depth DICOM scans into diagnostic-quality 8-bit representations. Designed a hybrid U-Net segmentation architecture with a ResNet-18 encoder and skip connections to preserve anatomical detail, achieving 98.42% validation pixel accuracy within 10 epochs. Adapted ImageNet-pretrained ResNet-18 models for skeletal age estimation and anatomical bounding box localization through continuous coordinate regression, achieving a localization Mean Absolute Error (MAE) of 0.0909.
- **GIS & Image Processing:** Leveraged QGIS to process and analyze multi-spectral Landsat-8 and Sentinel-2 satellite data, utilizing color composites to identify geographical features. Developed data plots and high-resolution animations of Hurricane Idalia's motion using NOAA GOES-16E imagery and ImageMagick. Conducted advanced spatial analysis through ArcGIS to evaluate the proximity impacts and environmental damage of California wildfires.
- **AI/ML Architecture:** Engineered a custom neural network from scratch, implementing a robust backpropagation algorithm to tune network parameters on synthetic datasets. Developed a comprehensive transfer learning benchmark to evaluate model performance across diverse domains, including computer vision (CV), natural language processing (NLP), and multi-modal integration.
- **MicroCaml:** Architected and engineered "MicroCaml," a functional sublanguage of OCaml. Developed a full pipeline including a custom Lexer for tokenization, a Parser for abstract syntax tree generation, and an Interpreter to execute higher-order functions. Additionally simulated a regular expression engine and a tree-based database system using OCaml.
- **Full-Stack Dev:** Built a responsive, multi-screen weather application in Kotlin featuring real-time API integration, multi-language support, and Google AdMob monetization. Developed a high-concurrency order processing supermarket application using a JavaScript/Node.js backend, a modern HTML/CSS frontend, and a MongoDB database for scalable JSON data storage.